Function Notes

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Definitions:

Image: $f(X) = \{\, f(x) \in B \mid x \in A \,\}$.

Pre-Image: $f^{-1}(X) = \{\, x \in X \mid f(x) \in B \,\}$.

Question 1:

$f([-3, 5]) = [3, 28]$, so $f^{-1}([3, 28]) = [-5, 5]$.

Question 2:

Consider the fucntion $f : A \to B$ and a subset $X \subseteq A$. Prove $X \subseteq f^{-1}(f(X))$.

Proof. Suppose $f : A \to B$ and $X \subseteq A$. Let $x \in X$. By definition of image, $f(x) \subseteq f(X)$. By definition of preimage, $x \subseteq f^{-1}(f(X))$. Since x is an arbitrary element of X, it follows that $X \subseteq f^{-1}(f(X))$. □

Question 3:

Proof. Assume that $y \in f(W \cap X)$. Then, there exists an $x \in W \cap X$ such that $y = f(x)$. Since $x \in W \cap X$, it follows that $x \in W$ and $x \in X$. By definition of image, $f(x) \in f(W)$ and $f(x) \in f(X)$. Therefore $y \in f(W) \cap f(X)$, and so $f(W \cap X) \subseteq f(W) \cap f(X)$. □